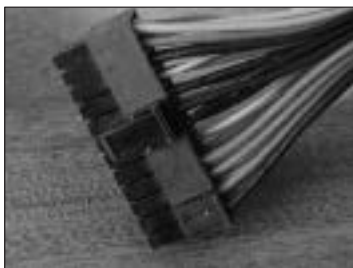


3.1.11 Power Connections

A few plug types seen in modern power units are:

○ 20 / 24-pin Motherboard Power Connector

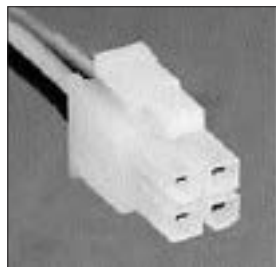
The 24-pin power connector adds four additional pins to the original 20-pin specification. Many SMPSEs offer a 24-pin plug with a detachable 4-pin section that can be used to plug in to older 20-pin sockets. The four additional pins provide additional power to the motherboard, and some motherboards with the 24-pin socket will not boot with the 20-pin plug.



A 24-pin motherboard power connector with four detachable pins

○ 4-pin 12 V Connector

Like the 24-pin plug, the 4-pin 12 V plug is needed to provide additional power to the motherboard. Booting will not be possible if this additional power is not provided to the motherboard.



A 12 V 4-pin connector

○ 4-Pin Molex Connector

Used to power hard disks, optical drives, cabinet fans, etc. Sometimes this may also be used to provide additional power to the graphics card, if it has the socket.



A 4-pin molex connector

○ 4-Pin Floppy Drive Connector

For the floppy drive, of course, if it's present.

○ 6-pin PCIE Connector

Powerful graphics cards require more power than the 75 watts that can be